



COURSE SLIDES

Agentic AI for Video Creation

Course Code: C436

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Course Trainer Profiles

1

Surendra Batukdeo

Emerging-technology, automation, data-intelligence, project, and technical-training experience.

2

Agus Salim

IT solutions, systems integration, cloud infrastructure, project, and cybersecurity experience.

3

Quah Chee Yong

Data science, Python, natural-language processing, machine-learning, and AI-training experience.

Let's Know Each Other

1 Introduce yourself and name one short-form video workflow you would like to improve.

2 Identify the stage that consumes the most time or creates the most rework.

3 State one video action you would never allow an agent to take without your approval.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

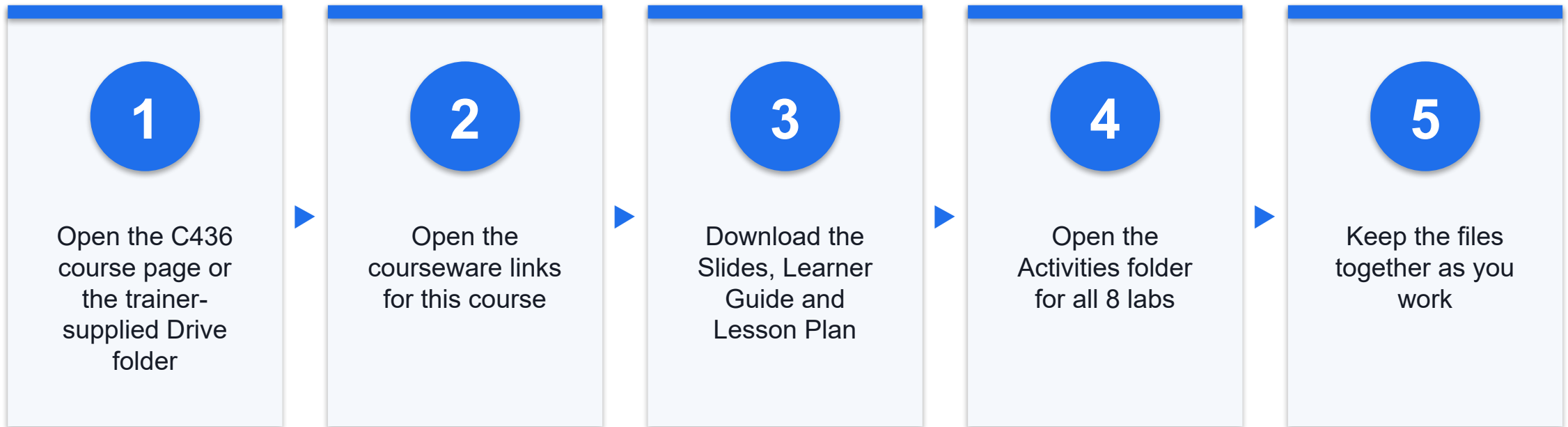
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Access Course Materials



Course page: <https://www.tertiarycourses.com.sg/agent-ai-for-video-creation.html>

Lesson Plan — 2 Days, 15 instructional hours

Course sequence

- Day 1 — Design the agent and generate an approved production pack
 - Topic 1: Getting Started with Agentic AI for Video (Labs 1–2)
 - Topic 2: Scripting and Generating Content with AI (Labs 3–4)

Day 2

- Day 2 — Assemble, review, orchestrate, and plan responsible scale
 - Topic 3: Editing and Assembling Videos with AI (Labs 5–6)
 - Topic 4: Automating and Scaling Video Production (Labs 7–8)
- Daily timing
 - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)

Learning Outcomes

1

Design the Agent

Explain agentic AI, select suitable video-production tools, write bounded prompts, and design a controlled end-to-end workflow.

2

Generate the Assets

Build AI-assisted research, scripting, storyboard, visual, voiceover, and music hand-offs from an approved creative brief.

3

Edit and Review

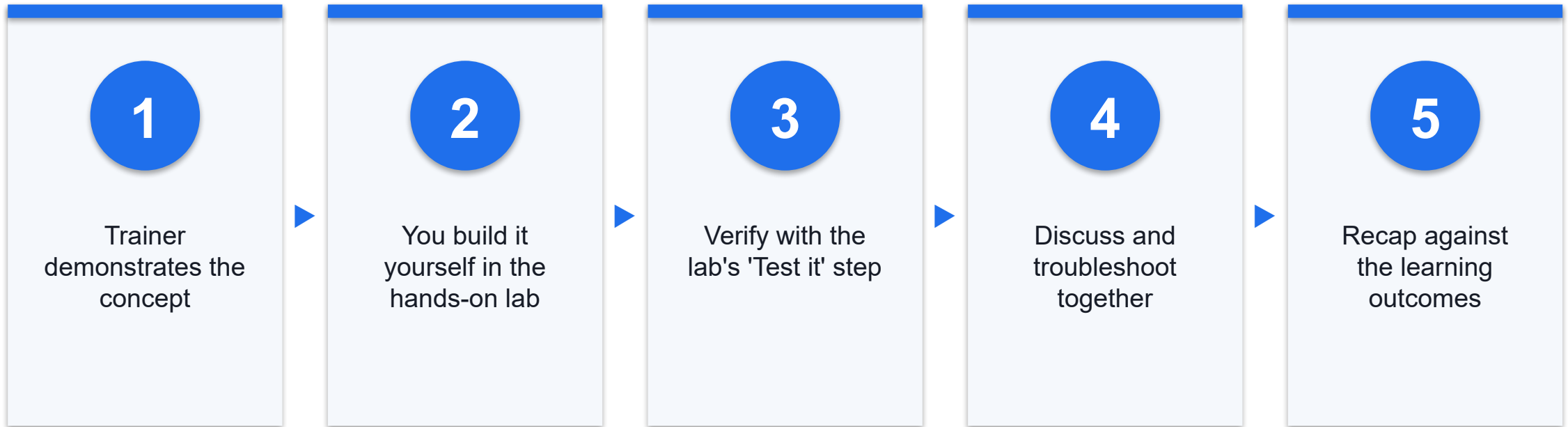
Assemble and refine a short-form video through automated editing, captioning, branding, and evidence-based quality checks.

4

Publish and Scale

Orchestrate multi-step video agents with human approval, prepare safe publishing actions, analyse performance, and plan responsible scale.

How You'll Learn





CORE CONCEPTS

Agentic Video Production Foundations

What Makes a Workflow Agentic

1

Goal and state

The system knows the desired outcome, accepted artifacts, current stage, and unresolved work.

2

Reasoning and routing

A model chooses among bounded next actions while deterministic rules enforce known constraints.

3

Tools and contracts

Every capability has typed inputs, portable outputs, permissions, budgets, and failure behavior.

4

Evidence and control

Logs, provenance, validators, approvals, and stop conditions keep the work reviewable.

The C436 Agent Loop

1

1 | Observe

Read the approved brief, state, manifest, and current evidence.

2

2 | Plan

Choose the next smallest action that advances a completion condition.

3

3 | Act

Call one allowed tool with structured inputs and bounded retries.

4

4 | Inspect

Validate the artifact, provenance, cost, and quality evidence.

5

5 | Decide

Accept, revise, retry within limits, or escalate to a named owner.

6

6 | Record

Persist the new state and an auditable event before continuing.

CORE PRINCIPLE

Autonomy is a design variable, not the goal.

Use deterministic automation for known operations, model judgment for bounded choices, and human approval for consequential actions.

The Connected Harbour Bean Build

Design

- Production contract
- Autonomy matrix
- Workflow map

Create

- Research register
- Timed script
- Asset manifest

Assemble

- Edit decision list
- Captions
- Vertical draft

How Every Lab Progresses

1

Start from an approved checkpoint and a named output contract.

2

Execute bounded steps using supplied synthetic data and placeholder secrets.

3

Run an observable 'Test It' check and record evidence.

4

Save the checkpoint so the next lab can continue without regeneration.

5

Reflect on which choices should remain deterministic, model-led, or human-approved.

Autonomy Matrix

1

Deterministic

Validation, naming, file movement, media probing, arithmetic, and schema checks.

2

Model-assisted

Idea clustering, script alternatives, shot suggestions, and evidence-linked issue drafts.

3

Human-approved

Claims, rights, likeness, brand exceptions, final edit, and release actions.

4

Prohibited by default

Unreviewed public posts, secret exposure, identity imitation, and unsupported claims.

The Production Contract

1

Purpose

Audience, channel, message, desired viewer action, and success decision.

2

Evidence

Approved facts, source register, unknowns, and permitted inference.

3

Creative constraints

Duration, aspect ratio, tone, brand, continuity, and prohibited content.

4

Operational limits

Tools, permissions, budgets, retries, owner, and finish condition.

Run Evidence

1

Identity

Run ID, job ID, artifact version, prompt version, and manifest checksum.

2

Actions

Tool name, input reference, output reference, time, status, and error.

3

Controls

Validation result, issue ID, approval scope, and decision owner.

4

Resources

Latency, generation count, cost estimate, and retry count.

TOPIC 01

Getting Started with Agentic AI for Video

Agentic AI foundations | Video, voice, and agent tools | Effective prompts | End-to-end workflow design

Key Concepts — Getting Started with Agentic AI for Video

1

Agent loop

A bounded cycle of observe, plan, act with tools, inspect evidence, and stop or escalate.

2

Production contract

A structured brief that fixes the audience, goal, source facts, constraints, deliverables, and approval gates.

3

Tool boundary

A named capability with explicit inputs, outputs, permissions, cost limits, and failure behavior.

4

Human control

People approve high-impact creative, rights, privacy, brand, and publishing decisions.

Introduction to Agentic AI for Video Creation

WHAT

- Agentic AI combines a model with instructions, tools, state, and a control loop so that the system can decide which bounded action to take next. In video production, an agent may inspect a brief, request missing facts, call research or generation tools, record outputs, check quality, and route an item for human approval. It is different from a one-shot chatbot because the workflow carries state and can take several tool-mediated steps toward a defined completion condition.

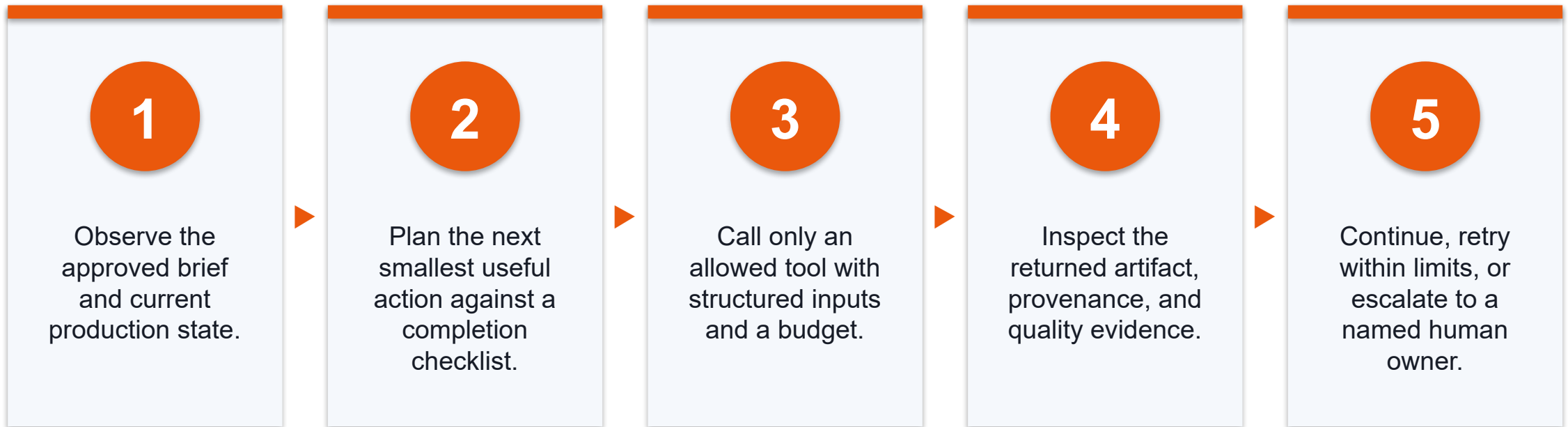
WHY IT MATTERS

- Video work contains creative uncertainty as well as operational dependencies. A reliable agent must know what it may decide, what evidence it must retain, and when it must stop. Treating every step as autonomous creates rights, privacy, cost, and brand risk; treating every step as fixed automation misses the value of reasoning. The practical design is bounded autonomy: deterministic rules for known operations, model judgment for well-framed choices, and human review for consequential actions.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Introduction to Agentic AI for Video Creation



In Practice — Introduction to Agentic AI for Video Creation

WORKED EXAMPLE

- Harbour Bean needs a 30-second vertical video about three ways to reduce bitter office coffee.
- A coordinator agent checks the brief, delegates research and scripting, then prepares scene requests.
- Every generator result is written to an asset register; publishing remains disabled until a person approves the final package.

DECISION GUIDE

- **Use when**
 - The task has several dependent steps and the next action depends on intermediate evidence.
 - Inputs, tool permissions, completion rules, and escalation paths can be stated clearly.
- **Avoid when**
 - The goal is vague, the source facts are unapproved, or there is no accountable owner.
 - The proposed action would publish, spend, or use a person's likeness without explicit review.

Quality Lens — Introduction to Agentic AI for Video Creation

Failure signal

- The system keeps generating options without reaching a defined finish state.

Repair move

- Add a completion checklist, iteration cap, and escalation rule.

Quality evidence

- The run log shows why each tool was called and who approved the release.

Popular AI Video, Voice, and Agent Tools

WHAT

- A production stack usually separates orchestration from specialist tools. The orchestrator holds the workflow state and routes tasks. Language models research and write. Image or video generators create candidate shots. Speech services produce narration. Editors and command-line media tools assemble, caption, mix, and export. Publishing and analytics APIs act only after credentials and permissions are configured.

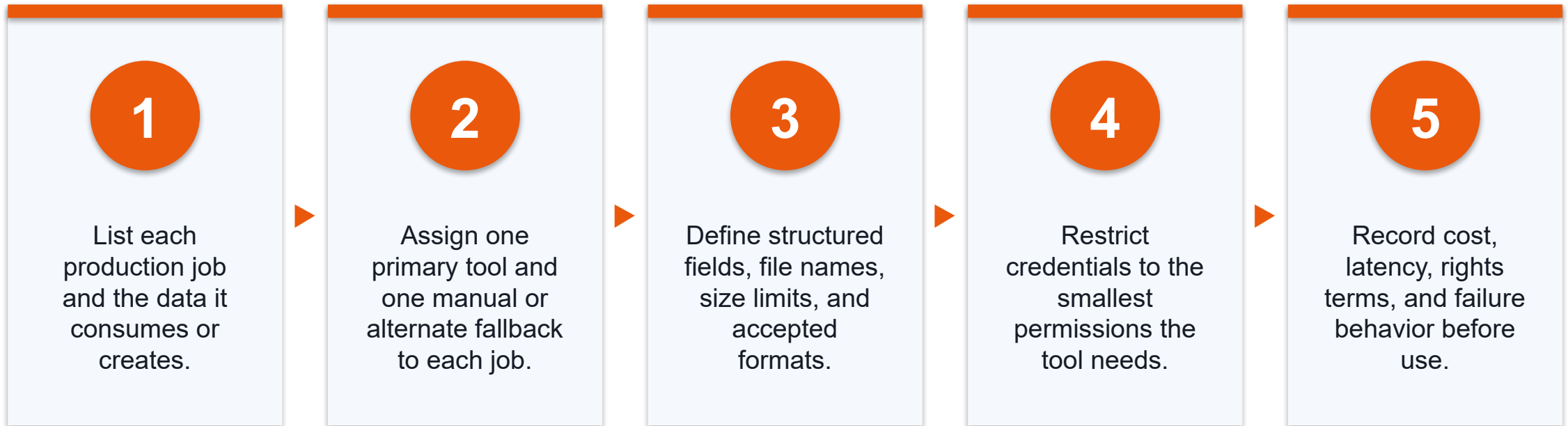
WHY IT MATTERS

- Product names and features change, but the jobs remain stable. A portable architecture defines each tool by capability, contract, and fallback instead of hiding the project inside one vendor. Structured outputs reduce broken hand-offs: a script agent should return timed beats, a visual tool should return a file plus provenance, and an editor should receive a validated manifest rather than free-form prose.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Popular AI Video, Voice, and Agent Tools



In Practice — Popular AI Video, Voice, and Agent Tools

WORKED EXAMPLE

- n8n coordinates eight stages without storing secret values in prompts.
- ChatGPT or Claude creates structured research and script outputs from approved source material.
- A chosen video generator supplies clips, ElevenLabs or an equivalent supplies narration, and FFmpeg or an editor assembles the vertical master.

DECISION GUIDE

- **Use when**
 - A specialist capability materially improves a defined stage and produces a portable output.
 - The team has an approved account, a fallback, and a clear rights and privacy position.
- **Avoid when**
 - A tool requires confidential material that is not approved for that service.
 - The workflow cannot export files, preserve provenance, or cap cost and retries.

Quality Lens — Popular AI Video, Voice, and Agent Tools

Failure signal

- A downstream stage cannot understand the previous tool's free-form response.

Repair move

- Define a JSON or file manifest contract with required fields.

Quality evidence

- Each stage can be swapped without redesigning the complete workflow.

Writing Effective Prompts for Video

WHAT

- An effective agent prompt is an operating contract, not a slogan. This course uses B-R-I-E-F: Background, Role and responsibility, Inputs and evidence, Execution constraints, and Format plus finish condition. The prompt states which facts are authoritative, which tools may be used, what must never happen, and exactly what valid output looks like.

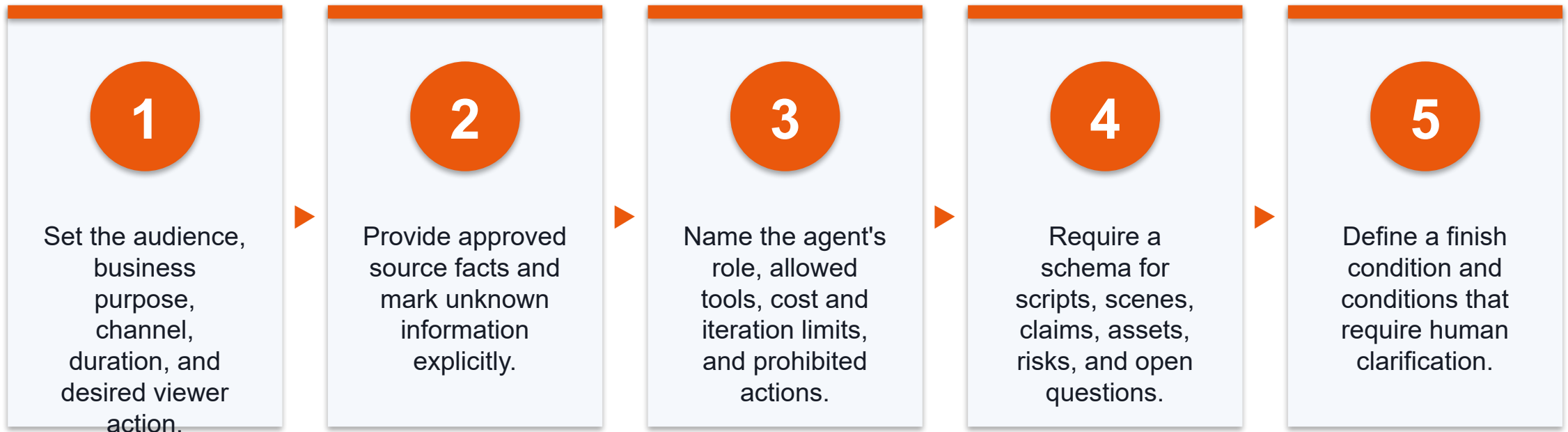
WHY IT MATTERS

- A creative request such as 'make a viral video' leaves the system to invent the audience, claims, style, and success rule. A structured prompt produces comparable alternatives, exposes missing evidence, and makes hand-offs machine-readable. Separate stable system instructions from per-run input, and validate important fields before a later tool acts on them.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Writing Effective Prompts for Video



In Practice — Writing Effective Prompts for Video

WORKED EXAMPLE

- Background: a Singapore cafe campaign for busy office workers.
- Inputs: only the supplied brand brief and audience signals may support claims.
- Execution: return three 25-35 second concepts; do not publish or imitate a living creator.
- Format: valid JSON with hook, timed beats, scene list, evidence, risks, and status.

DECISION GUIDE

- **Use when**
 - Inputs and desired outputs can be bounded and checked.
 - Several tools or agents need a common production contract.
- **Avoid when**
 - The instruction hides a policy decision that an accountable person must make.
 - The output cannot be validated before a costly or external action follows.

Quality Lens — Writing Effective Prompts for Video

Failure signal

- The output looks plausible but omits required fields or invents a source.

Repair move

- Add a schema, evidence citations, explicit unknown handling, and a validator.

Quality evidence

- A second run can use the same prompt contract and produce structurally valid output.

Designing an End-to-End Video Agent Workflow

WHAT

- A workflow turns the production contract into observable stages: intake, research, concept selection, script and storyboard, asset generation, assembly, quality review, approval, publishing preparation, and learning. Each stage receives a defined input, writes a durable output, and returns a status such as ready, needs-revision, blocked, or approved.

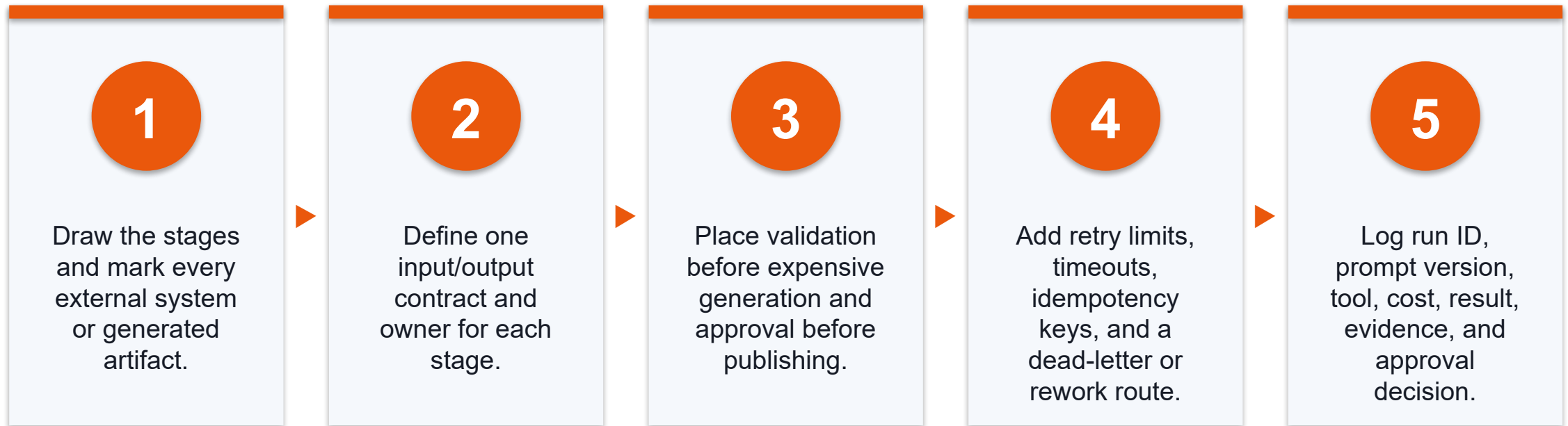
WHY IT MATTERS

- Large all-in-one agents are difficult to debug and may repeat expensive work. Stage boundaries create checkpoints, allow deterministic validation, and let a learner restart from the last accepted artifact. Idempotency keys prevent a retry from duplicating external actions; run identifiers connect every artifact and log entry to the same job.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Designing an End-to-End Video Agent Workflow



In Practice — Designing an End-to-End Video Agent Workflow

WORKED EXAMPLE

- Run HB-001 advances only when `brief_status=approved` and `script_status=approved`.
- Scene generation retries once for a technical failure but routes a rights concern to human review.
- The publishing node receives a release package only; it cannot see raw research or use an unapproved file.

DECISION GUIDE

- **Use when**
 - The work has repeatable stages and accepted artifacts can be reused.
 - The system can save state and resume safely after a failure.
- **Avoid when**
 - The process is a one-off creative conversation with no reusable structure.
 - A retry might duplicate an external action and no idempotency or approval control exists.

Quality Lens — Designing an End-to-End Video Agent Workflow

Failure signal

- A failed final step forces the whole production to regenerate.

Repair move

- Persist accepted stage outputs and resume from a checkpoint.

Quality evidence

- A run can stop, rejoin, and explain its complete state without guesswork.

Hands-On Labs — Getting Started with Agentic AI for Video

Lab 1

- Design the Video Agent Production Contract and Autonomy Matrix

Lab 2

- Build and Run the Bounded Video Planning Agent in n8n

Design the Video Agent Production Contract and Autonomy Matrix

LAB 1

You will turn the synthetic Harbour Bean campaign brief into a structured production contract, classify each task by autonomy level, and mark the evidence, approval, retry, and stop rules for the workflow.

You'll build

A completed production-contract.json and autonomy-matrix.csv for run HB-001, saved in the connected project checkpoint.

1 Inspect the synthetic brief

2 Define the production contract

3 Classify task autonomy

4 Validate the files

Tools: Text editor, supplied synthetic brief, JSON validator

Design the Video Agent Production Contract and Autonomy Matrix

Test it

The validator must print LAB-01 PASS. It enforces JSON types, positive budget, all nine autonomy rows, prohibition of public publishing, and prohibition of secret exposure. The placeholder search must return no matches.

Build and Run the Bounded Video Planning Agent in n8n

LAB 2

You will import a supplied n8n workflow, trace its trigger, validation, planning, guardrail, and output stages, then run one ready case and one blocked case. The agent uses deterministic mock planning so everyone can verify the control pattern without a paid API.

You'll build

An n8n workflow named C436-HB-001-Planning-Agent plus exported ready and blocked execution evidence.

1 Import the workflow

2 Inspect contracts and gates

3 Run the ready job

4 Force a blocked job

Tools: n8n, supplied workflow JSON, production contract

Build and Run the Bounded Video Planning Agent in n8n

Test it

The Lab 2 validator must print LAB-02 PASS after reading ready-run.json, blocked-run.json, and ready-restored-run.json. Both ready results must remain non-publishing; the controlled missing-evidence result must fail closed.

Recap — Getting Started with Agentic AI for Video

1

You can now: LO1: explain agentic AI and design a controlled video workflow with explicit human decision points

2

You can now: LO1: use structured prompts and tool boundaries to implement a resumable planning workflow

TOPIC 02

Scripting and Generating Content with AI

Trend and idea research | Scripts and storyboards | Visuals and B-roll | Voiceovers and music

Key Concepts — Scripting and Generating Content with AI

1

Source register

A dated record of evidence, relevance, permissions, and claims that may be used.

2

Timed beat sheet

A sequence linking narration, visual, on-screen text, sound, and duration.

3

Asset manifest

The authoritative list of scene files, prompts, versions, rights notes, and status.

4

Continuity bible

Stable character, product, lighting, palette, camera, and negative constraints across generations.

Researching Trends and Ideas

WHAT

- Agentic research is a bounded evidence-gathering workflow. It starts with a question and approved source types, retrieves observations, records provenance, separates evidence from inference, and produces candidate ideas that are traceable to audience need. Trend signals are clues for timing and format, not permission to copy another creator.

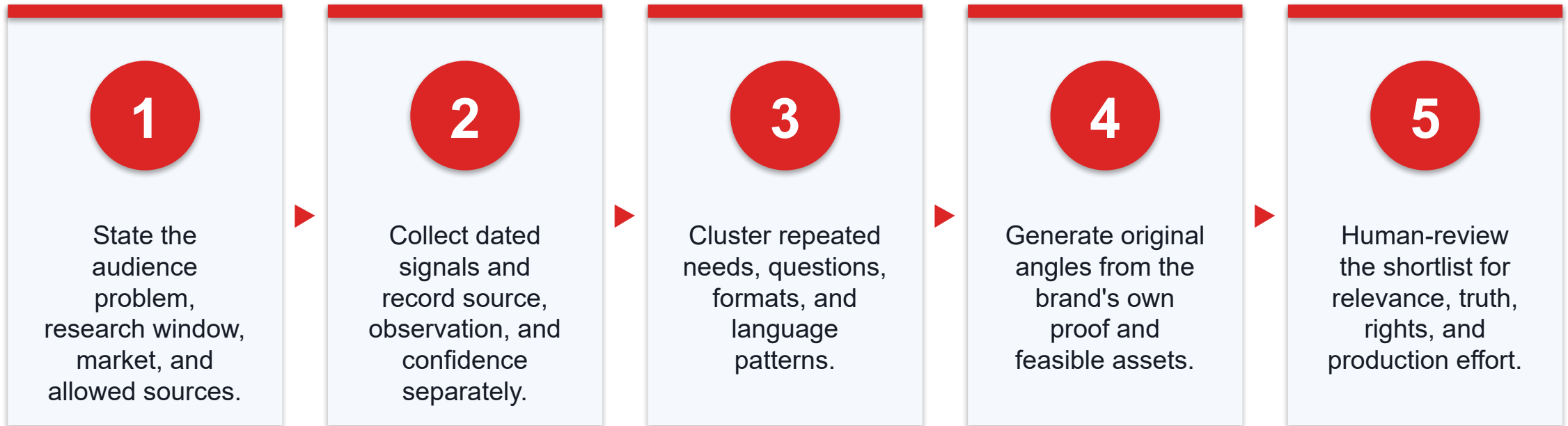
WHY IT MATTERS

- A research agent can gather many examples quickly, but search results may be stale, duplicated, promotional, or detached from the target audience. A source register and evidence threshold prevent a script agent from treating popularity as truth. The output should include open questions and rejected ideas, not only a polished recommendation.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Researching Trends and Ideas



In Practice — Researching Trends and Ideas

WORKED EXAMPLE

- Audience signals mention bitter office coffee, inconsistent scoops, and limited time.
- The agent groups these into three teachable variables instead of copying a trending cafe video.
- Each candidate idea cites the supplied observation and the brand fact that supports it.

DECISION GUIDE

- **Use when**
 - The question, time window, source types, and evidence threshold are explicit.
 - A person can inspect the source register before ideas move to scripting.
- **Avoid when**
 - The workflow would scrape personal data or bypass platform access controls.
 - Trend volume is being used as proof for a product or health claim.

Quality Lens — Researching Trends and Ideas

Failure signal

- The idea bank contains unsourced claims and near-copies of examples.

Repair move

- Require provenance, originality notes, and a reason for every retained idea.

Quality evidence

- The chosen idea can be traced to audience evidence and approved brand facts.

Generating Scripts and Storyboards

WHAT

- A script translates the selected promise into spoken words and on-screen text; a storyboard translates the same promise into timed visual evidence. The agent should work in beats rather than a single paragraph. Every beat states time, narration, picture, overlay, sound, claim source, and transition so production and review share one model.

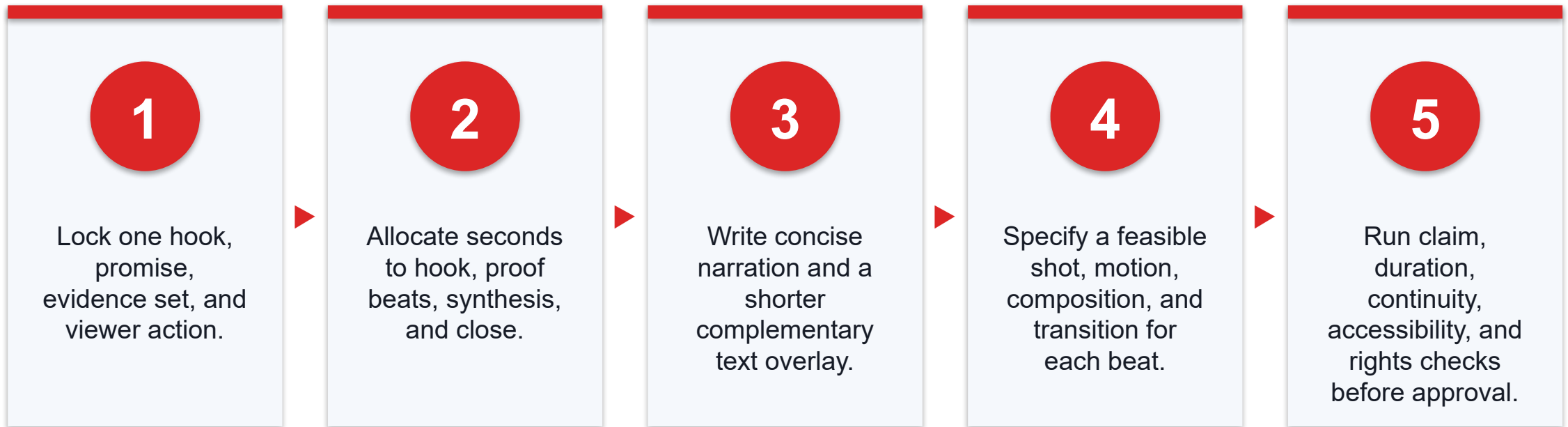
WHY IT MATTERS

- Short-form videos fail when words, visuals, and captions compete or when the final edit exceeds the target duration. Time budgeting before generation constrains scope and avoids unnecessary media cost. A storyboard also exposes scenes that cannot be produced safely or consistently before the system requests them.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Generating Scripts and Storyboards



In Practice — Generating Scripts and Storyboards

WORKED EXAMPLE

- A 30-second script allocates 3 seconds to the hook, 21 seconds to three fixes, and 6 seconds to recap and close.
- The spoken line explains the fix while on-screen text labels only the variable.
- Every claim points back to an approved source row; the storyboard marks one synthetic scene for disclosure.

DECISION GUIDE

- **Use when**
 - The message and source facts are approved and a duration can be fixed.
 - The next stage needs structured scene requests and voice text.
- **Avoid when**
 - The script requires unsupported before-and-after proof or an unapproved likeness.
 - The storyboard depends on complex continuity the chosen generator cannot maintain.

Quality Lens — Generating Scripts and Storyboards

Failure signal

- Narration, captions, and visuals repeat the same sentence.

Repair move

- Assign a distinct job to each channel: explain, label, or demonstrate.

Quality evidence

- A cold reader can produce the intended cut from the timed beat sheet.

Creating Visuals and B-Roll with AI

WHAT

- Visual generation converts storyboard beats into candidate images or clips. A strong scene request defines subject, action, environment, composition, camera, lighting, palette, duration, and negative constraints. The asset manifest records the prompt, model or tool, date, version, rights note, and accepted use for every file.

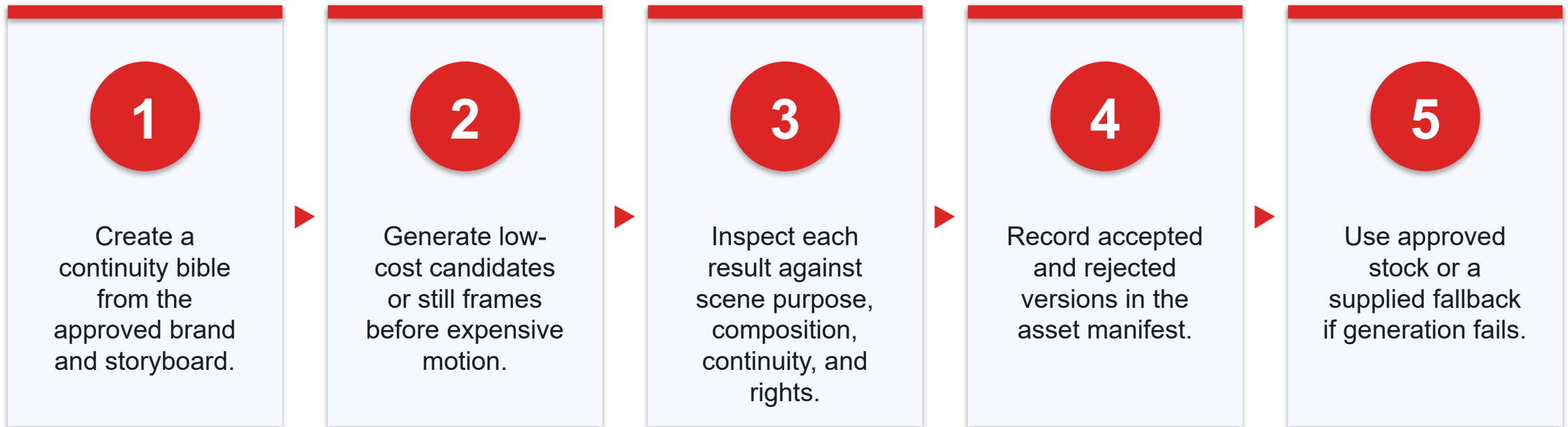
WHY IT MATTERS

- One attractive clip does not create a coherent video. Continuity, editability, and factual fit matter more than isolated novelty. Generate short modular shots, preserve safe areas for vertical overlays, and review anatomy, text, logos, product details, and motion before a clip is accepted.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Creating Visuals and B-Roll with AI



In Practice — Creating Visuals and B-Roll with AI

WORKED EXAMPLE

- Scene 02 shows a measured scoop and timer on a clean office pantry counter.
- The prompt fixes the Harbour Bean palette and leaves the lower third clear for captions.
- A clip with a malformed product label is rejected even if its motion is appealing.

DECISION GUIDE

- **Use when**

- The scene is synthetic, clearly bounded, and does not require a real person's identity.
- The team can retain provenance and has a fallback for failed generation.

- **Avoid when**

- The scene impersonates a person, fabricates a real event, or uses protected material without permission.
- The required text or exact product geometry should be produced deterministically in the editor.

Quality Lens — Creating Visuals and B-Roll with AI

Failure signal

- Accepted scenes vary in lighting, product form, and screen direction.

Repair move

- Reuse a continuity bible and generate by shot family with reference frames where allowed.

Quality evidence

- Every accepted file has provenance and supports a named storyboard beat.

Generating Voiceovers and Music

WHAT

- Voice generation turns approved narration into timed audio. Music supports pace and emotion without masking speech. The workflow selects a permitted voice, normalises written text for speech, produces an audio file, checks pronunciation and duration, and records the service, settings, consent, and usage terms.

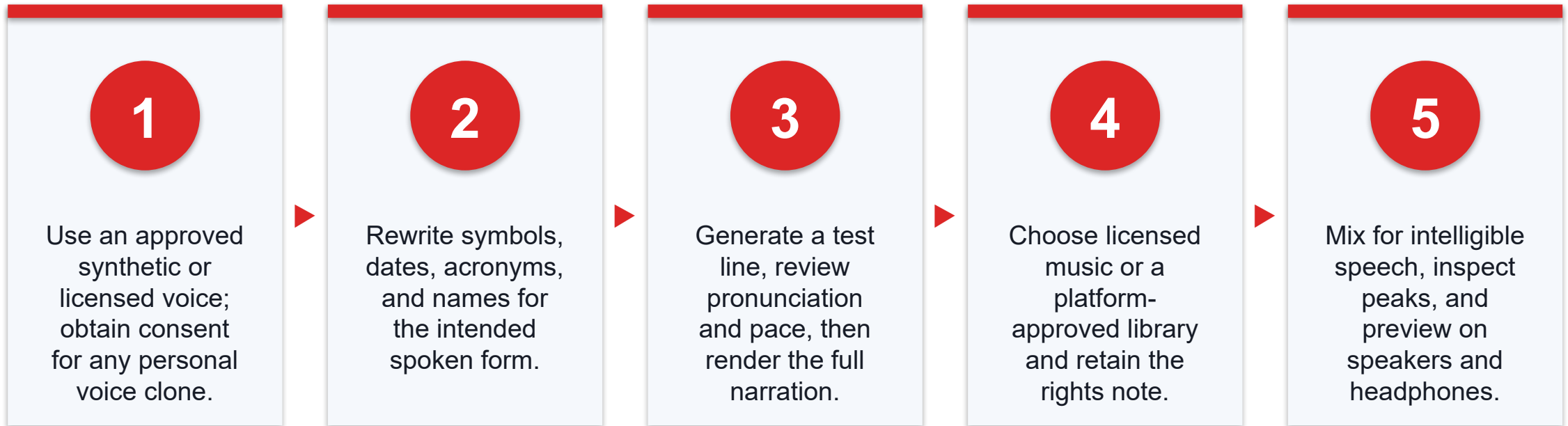
WHY IT MATTERS

- Audio can make a visually strong video inaccessible or untrustworthy. Unauthorised voice cloning, poor pronunciation, excessive loudness, and unclear licensing create avoidable risk. Voice and music are therefore separate reviewed assets with explicit owners and fallback options.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Generating Voiceovers and Music



In Practice — Generating Voiceovers and Music

WORKED EXAMPLE

- The agent converts '3 fixes in 30 sec' to natural spoken wording before synthesis.
- The narrator file is saved with the script version and chosen settings.
- Background music is ducked under speech and the no-music export remains available as a fallback.

DECISION GUIDE

- **Use when**
 - The voice identity and music rights are explicit and suitable for the intended channel.
 - A person can listen to and approve the complete audio before assembly.
- **Avoid when**
 - The workflow clones a voice without documented permission.
 - Licensing, commercial use, or territorial rights cannot be established.

Quality Lens — Generating Voiceovers and Music

Failure signal

- The narration sounds natural but mispronounces the brand or exceeds the scene timing.

Repair move

- Add a pronunciation dictionary, rewrite for speech, and regenerate only the affected line.

Quality evidence

- The approved audio matches the timed script and has a recorded rights basis.

Hands-On Labs — Scripting and Generating Content with AI

Lab 3

- Run the Research-to-Script Agent and Approve a Timed Storyboard

Lab 4

- Build the Visual, Voiceover, and Music Asset Request Pack

Run the Research-to-Script Agent and Approve a Timed Storyboard

LAB 3

You will use an approved AI assistant with a structured B-R-I-E-F prompt, or the supplied offline result, to cluster synthetic audience signals, propose distinct concepts, and return a timed script and storyboard. You will validate the schema and manually approve only evidence-linked content.

You'll build

A source-register.csv, research-script-prompt.txt, script-storyboard.json, captions-script-v1.vtt, and storyboard-approval.md in stage folder 02-create.

1 Register evidence

2 Run a structured prompt

3 Validate timed output

4 Review claims and scenes

Tools: Approved AI assistant or offline fallback, text editor, supplied synthetic data

Run the Research-to-Script Agent and Approve a Timed Storyboard

Test it

The validator must print LAB-03 PASS. It enforces run ID, script_ready status, exact Lab 2 fingerprint hand-off, human approval token, three concepts, required beat fields/source IDs, contiguous timing, and captions.

Build the Visual, Voiceover, and Music Asset Request Pack

LAB 4

You will import a deterministic asset-request workflow, generate a continuity bible and one request per storyboard beat, prepare narration and music briefs, and review the resulting asset manifest. Optional live generation is kept outside the required path; supplied placeholder media lets everyone continue.

You'll build

A continuity-bible.json, asset-requests.json, narration.txt, music-brief.md, and asset-manifest.csv with approved placeholders or authorised generated files.

1 Import the request agent

2 Generate continuity and requests

3 Prepare voice and music briefs

4 Register files and provenance

Tools: n8n, optional approved video/voice tools, supplied placeholder media

Build the Visual, Voiceover, and Music Asset Request Pack

Test it

The validator must print LAB-04 PASS. It requires the exact S01-S05, A01, and A02 set, correct types, accepted approved-rights status, files, unique IDs, and the human APPROVED_FOR_ASSEMBLY token.

Recap — Scripting and Generating Content with AI

1

You can now: LO2: create evidence-linked research, script, caption, and storyboard hand-offs from an approved brief

2

You can now: LO2: create structured media-generation requests with continuity, provenance, rights, and fallback controls

TOPIC 03

Editing and Assembling Videos with AI

Automated editing | Captions, effects, and branding | Short-form assembly | AI-supported review and refinement

Key Concepts — Editing and Assembling Videos with AI

1

Edit decision list

A deterministic map from source assets to timeline order, in/out points, overlays, and transitions.

2

Media probe

Machine-readable evidence about codec, resolution, frame rate, duration, audio, and streams.

3

Caption track

Time-aligned text stored as a portable file and optionally burned into the picture.

4

Quality gate

A documented set of technical, editorial, accessibility, rights, and brand checks.

Automating Video Editing

WHAT

- Automated editing turns an approved manifest and edit decision list into a repeatable render. Deterministic operations such as trim, scale, crop, concatenate, overlay, caption, and audio mix are best handled by an editor or media pipeline. A model may propose the sequence or repair plan, but the render command should be explicit and logged.

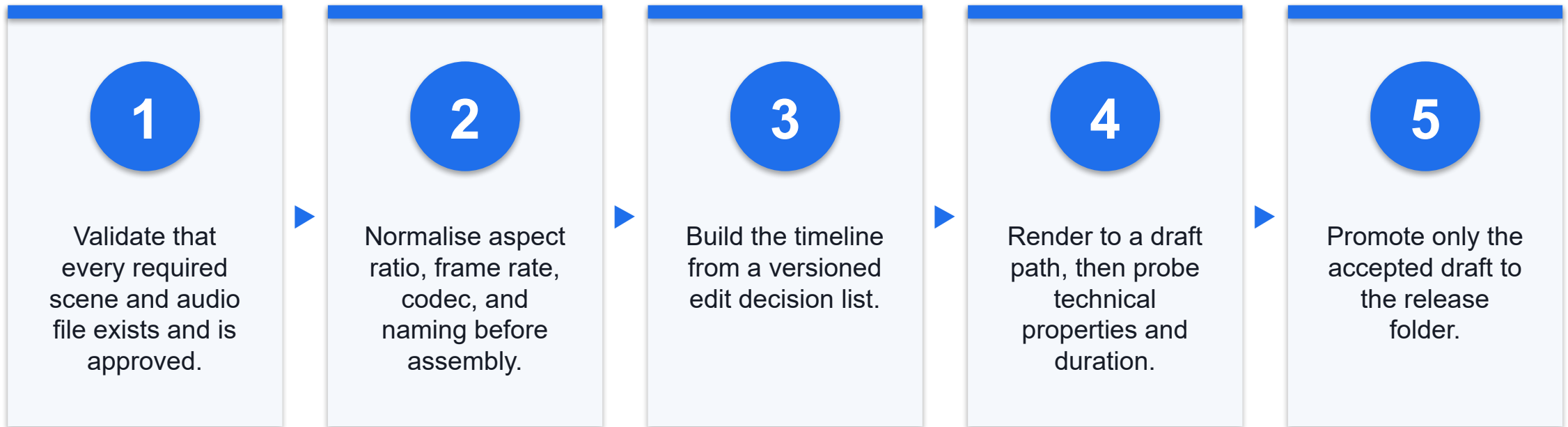
WHY IT MATTERS

- Regenerating a full timeline for a small change wastes time and can introduce new errors. Manifest-driven assembly makes each input and transformation visible, supports selective reruns, and produces the same output from the same accepted files. A media probe before and after the render catches mismatched dimensions, missing audio, and duration drift.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Automating Video Editing



In Practice — Automating Video Editing

WORKED EXAMPLE

- The manifest lists five vertical scenes, one narration track, music, and a caption file.
- The assembly script scales to 1080x1920, applies bounded trims, mixes audio, and exports a draft.
- A duration check routes a 33.8-second result back because the contract requires no more than 32 seconds.

DECISION GUIDE

- **Use when**
 - The repeated edit can be expressed through a stable manifest or template.
 - The team needs reproducible renders and traceable revisions.
- **Avoid when**
 - The creative decision depends on subtle performance judgment that is not encoded in the plan.
 - The source media cannot be legally or technically processed by the chosen tool.

Quality Lens — Automating Video Editing

Failure signal

- A rerun changes unrelated parts of the video.

Repair move

- Pin inputs, settings, and the edit decision list to a run version.

Quality evidence

- The render log and media probe match the declared release specification.

Adding Captions, Effects, and Branding

WHAT

- Captions represent speech and essential audio in time-aligned text. Branding uses controlled typography, colour, logo placement, and tone. Effects should guide attention or clarify change; they are not a substitute for a coherent story. Separate caption text from styling so the same approved content can be exported as WebVTT, platform captions, or burned-in text.

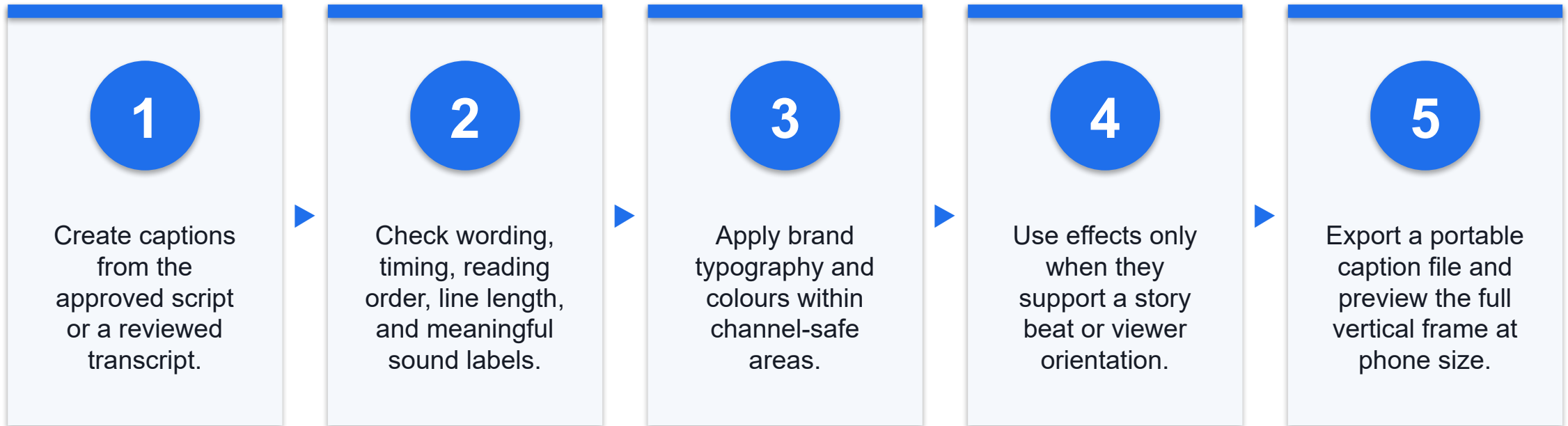
WHY IT MATTERS

- Most short-form video is watched in varied sound and attention conditions. Accurate captions improve access and comprehension, while a consistent safe-area layout prevents text and controls from colliding. Automated transcription is a draft: names, numbers, timing, line breaks, and speaker meaning still require human review.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Adding Captions, Effects, and Branding



In Practice — Adding Captions, Effects, and Branding

WORKED EXAMPLE

- The WebVTT file begins with the required header and contains ordered cue timings.
- On-screen keywords complement rather than duplicate the complete caption line.
- The logo and lower-third remain clear of common interface overlays.

DECISION GUIDE

- **Use when**

- The transcript can be reviewed against the final audio.
- Brand assets and usage rules are approved and available.

- **Avoid when**

- The system guesses inaudible speech or decorative text hides the subject.
- A generated logo, typeface, or sound effect has uncertain rights.

Quality Lens — Adding Captions, Effects, and Branding

Failure signal

- Captions are accurate but unreadable on a phone or out of sync after an edit.

Repair move

- Regenerate timings from the final audio and recheck safe area, contrast, and line breaks.

Quality evidence

- The final file passes text, timing, contrast, and mobile-preview checks.

Assembling Short-Form Videos

WHAT

- Assembly is the editorial act of making every visual, spoken line, caption, and sound serve one promise. The hook establishes relevance, the body delivers proof through a clear sequence, and the close completes the promise with a proportionate next action. Rhythm comes from information change, not from arbitrary rapid cuts.

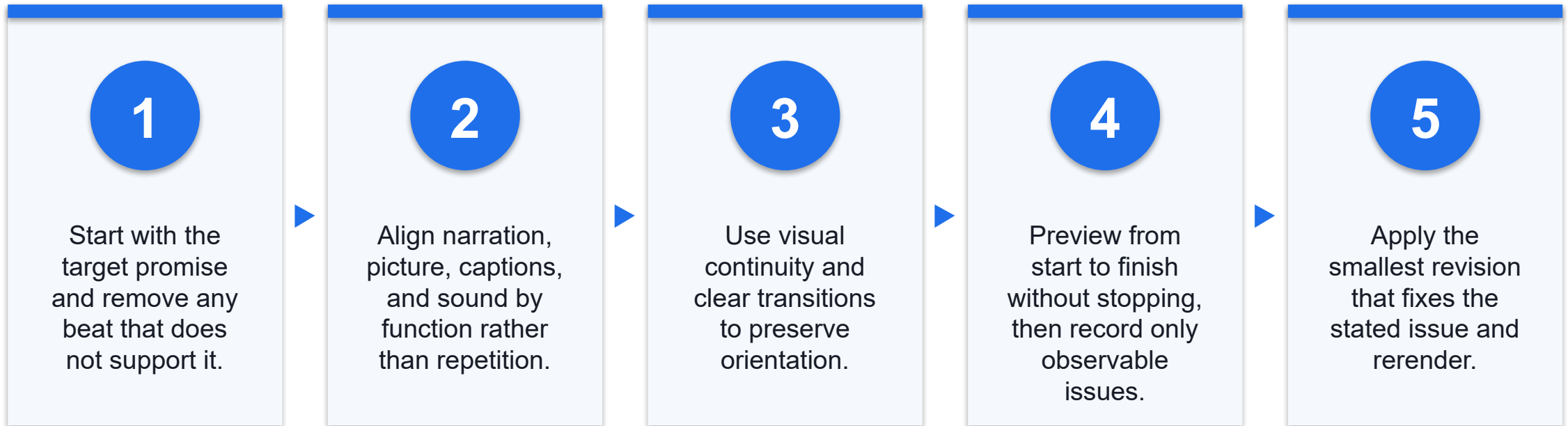
WHY IT MATTERS

- An agent can detect missing files, long gaps, repeated shots, or timing mismatches, but it cannot own the final communication judgment. The creator must watch the complete video as a viewer, with sound on and off, and confirm that the story remains understandable, truthful, and appropriately paced.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Assembling Short-Form Videos



In Practice — Assembling Short-Form Videos

WORKED EXAMPLE

- The opening shows the bitter-cup problem while narration names the viewer situation.
- Three proof beats demonstrate variables in the same order as the spoken explanation.
- The close summarises the checklist and invites the viewer to save it.

DECISION GUIDE

• Use when

- The script, storyboard, media, and audio are approved enough for a complete draft.
- The team can preview on the intended aspect ratio and device.

• Avoid when

- Essential evidence is missing and the edit would disguise that gap.
- The workflow optimises only for cut frequency or novelty.

Quality Lens — Assembling Short-Form Videos

Failure signal

- The edit is energetic but the viewer cannot restate the three fixes.

Repair move

- Restore causal order and remove decorative elements that compete with proof.

Quality evidence

- A cold viewer identifies the promise, proof, and next action without explanation.

Reviewing and Refining with AI

WHAT

- A review agent inspects the draft against a declared rubric and returns evidence, severity, location, and a bounded repair suggestion. It may compare the script to captions, probe the file, detect missing manifest entries, or flag brand and rights questions. It does not give itself permission to approve or publish its own work.

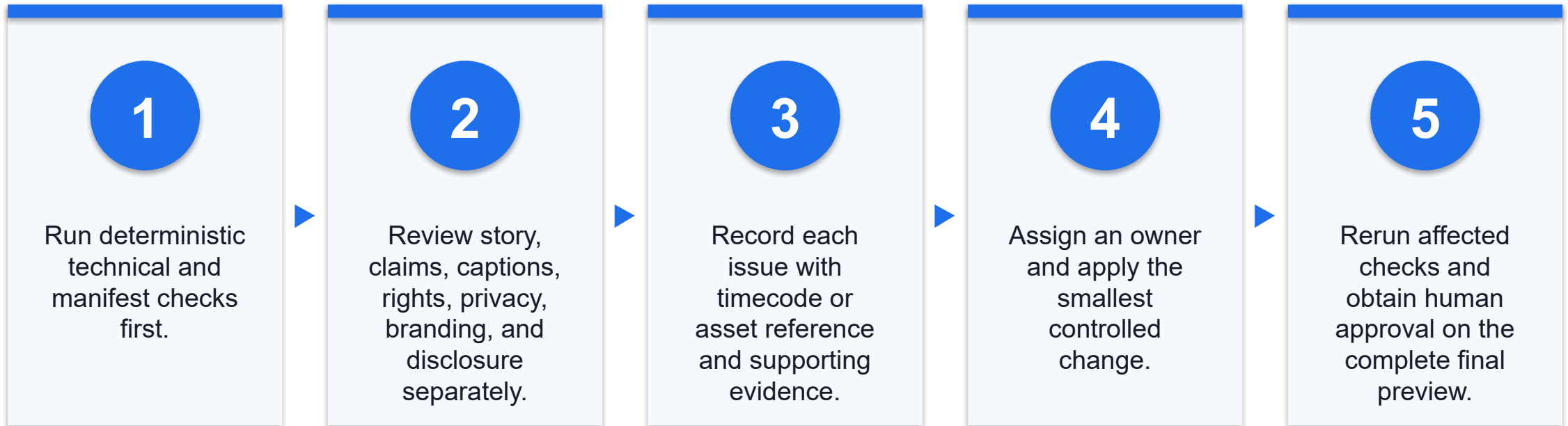
WHY IT MATTERS

- Unstructured feedback such as 'make it more engaging' causes uncontrolled rewrites. An issue register turns observations into reproducible decisions: issue ID, category, evidence, severity, owner, fix, and recheck result. Independent checks reduce the risk that the same assumptions survive from generation into review.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Reviewing and Refining with AI



In Practice — Reviewing and Refining with AI

WORKED EXAMPLE

- The agent flags a caption mismatch at 00:12, a missing rights note for asset S03, and a 1.8-second duration overrun.
- The editor fixes only those items and links the new render to the same run.
- A person confirms the whole video after the automated checks return clear.

DECISION GUIDE

- **Use when**
 - The rubric, evidence sources, and severity thresholds are explicit.
 - A human remains accountable for ambiguous creative and release decisions.
- **Avoid when**
 - The reviewer shares the same hidden context and merely confirms its own output.
 - A score is used without evidence, location, or a repair path.

Quality Lens — Reviewing and Refining with AI

Failure signal

- The system gives a high quality score while required release evidence is missing.

Repair move

- Use blocking gates for mandatory fields and evidence-linked findings for judgment.

Quality evidence

- Every cleared issue has a recorded recheck and the final approval names a person.

Hands-On Labs — Editing and Assembling Videos with AI

Lab 5

- Assemble and Probe the Captioned Vertical Video

Lab 6

- Run the Independent Video Review Gate and Repair One Finding

Assemble and Probe the Captioned Vertical Video

LAB 5

You will use the supplied PowerShell and FFmpeg assembly path to verify the accepted manifest sources, create deterministic scene clips, add narrated audio and a muxed WebVTT-derived caption track. You will then probe the draft and save machine-readable technical evidence.

You'll build

A vertical-draft-v1.mp4, captions-v1.vtt, edit-decision-list.json, ffprobe-v1.json, render-log-v1.txt, and render-evidence-v1.json in stage folder 03-edit.

1 Validate inputs

2 Build edit decision list

3 Render vertical draft

4 Attach caption evidence

Tools: PowerShell, FFmpeg, FFprobe, supplied assembly script

Assemble and Probe the Captioned Vertical Video

Test it

The validator must print LAB-05 PASS. It hashes and inspects the actual MP4, probe, captions, and log; asserts H.264, 1080x1920, 30 fps, yuv420p, 30-second duration, audio/subtitles, ordered VTT, and approval.

Run the Independent Video Review Gate and Repair One Finding

LAB 6

You will import an n8n review workflow that reads exact technical, caption, and manifest evidence, validates 28–32 seconds plus ordered contiguous cues, creates blocking and advisory findings, and never self-approves. You will inject a controlled defect, verify that the gate blocks, repair the data, rerun, and record the final human decision after watching the complete draft.

You'll build

A review-agent workflow export, issue-register-v1.json, issue-register-v2.json, repair-log.md, and final-review-approval.json.

1 Import independent review

2 Inject a controlled defect

3 Verify the blocking gate

4 Repair and recheck

Tools: n8n, FFprobe evidence, text editor, local media player

Run the Independent Video Review Gate and Repair One Finding

Test it

The validator must print LAB-06 PASS: rights, invalid duration, or bad caption timing must block; repaired evidence stays non-releasing; the human JSON approval must bind the actual video hash and private-only scope.

Recap — Editing and Assembling Videos with AI

1

You can now: LO3: assemble a reproducible short-form video with captions, branding, and technical evidence

2

You can now: LO3: review and refine a video through technical, editorial, accessibility, brand, rights, and release gates

TOPIC 04

Automating and Scaling Video Production

Multi-step video agents | Publishing and scheduling | Performance analysis | Scalable content pipelines

Key Concepts — Automating and Scaling Video Production

1

Orchestrator

The workflow component that routes state between specialist stages and enforces gates.

2

Approval token

A recorded, scoped decision that authorises one release action for one approved package.

3

Metric contract

A definition of the decision, grain, window, numerator, denominator, and exclusions for each KPI.

4

Scaling guardrail

A limit on volume, spend, retries, permissions, or variance that grows with automation.

Building Multi-Step Video Agents

WHAT

- A multi-step video system may use a coordinator plus specialist research, script, asset, edit, review, and release stages. Specialisation is valuable when each role has a distinct tool set and output contract. The coordinator should route state and enforce policies, not rewrite every artifact.

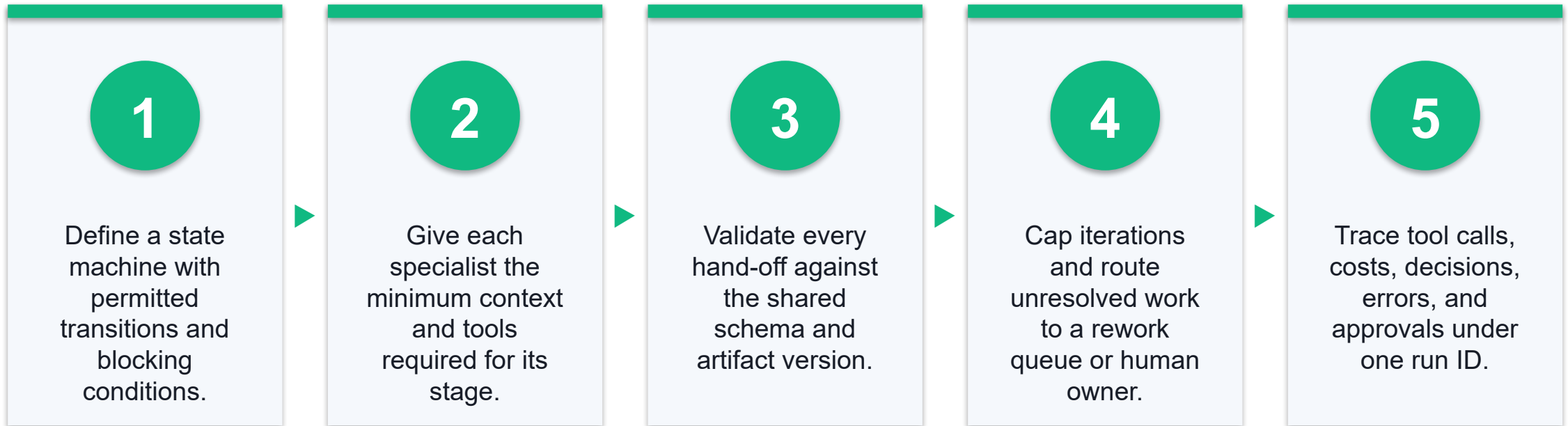
WHY IT MATTERS

- Adding agents increases hand-offs, cost, latency, and opportunities for inconsistent assumptions. Start with one workflow and split a stage only when the separation improves control, parallel work, specialist tooling, or evaluation. Shared run state must distinguish approved artifacts from drafts.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Building Multi-Step Video Agents



In Practice — Building Multi-Step Video Agents

WORKED EXAMPLE

- HB-001 moves `brief_approved` -> `script_ready` -> `assets_ready` -> `draft_ready` -> `release_ready`.
- A rights flag prevents the release stage from running even when all media files exist.
- The coordinator sends the issue back to the asset owner rather than regenerating the script.

DECISION GUIDE

- **Use when**

- Specialist roles have distinct contracts, tools, or review criteria.
- The orchestrator can persist state and enforce permitted transitions.

- **Avoid when**

- One prompt and one tool would solve the bounded task more reliably.
- Agents share unrestricted credentials or can silently overwrite accepted artifacts.

Quality Lens — Building Multi-Step Video Agents

Failure signal

- Agents loop or disagree about which artifact is current.

Repair move

- Use a state machine, immutable versions, and one authoritative manifest.

Quality evidence

- The complete run can be reconstructed from state transitions and logs.

Publishing and Scheduling Across Platforms

WHAT

- Publishing is an external, high-impact action. The release stage should receive only an approved video, captions, metadata, disclosure decision, rights record, target account, privacy setting, and schedule. The default learning path produces a dry-run request and keeps visibility private until an authorised person reviews platform-specific fields and consents.

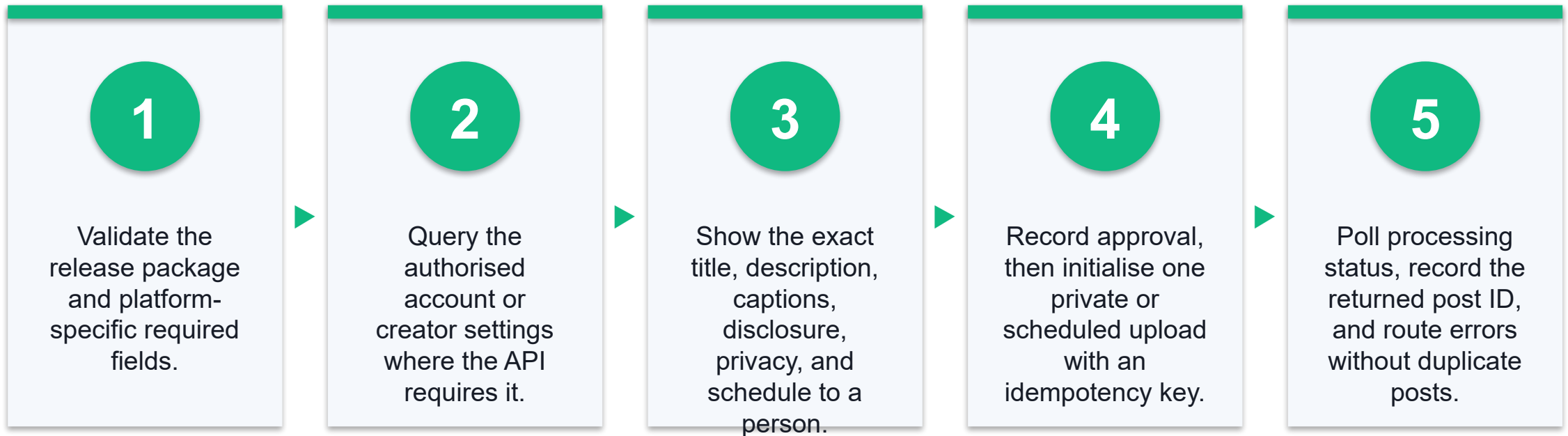
WHY IT MATTERS

- Platforms have different permissions, quotas, disclosure fields, and audit requirements. Automating the final click without checking creator information, audience settings, privacy, and synthetic-media disclosure can cause irreversible mistakes. A scoped approval token and idempotency key ensure one approved package produces at most one intended post.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Publishing and Scheduling Across Platforms



In Practice — Publishing and Scheduling Across Platforms

WORKED EXAMPLE

- The C436 lab creates a private dry-run package for YouTube and a TikTok request preview.
- The release remains blocked until disclosure, rights, and owner_approval fields are complete.
- The workflow records the intended target and package checksum before any live integration is enabled.

DECISION GUIDE

- **Use when**

- The account owner has authorised the app and can preview every required field.
- Private or draft mode, error handling, and duplicate prevention are available.

- **Avoid when**

- The workflow would post publicly without explicit consent and a complete preview.
- Credentials, disclosure, rights, or platform audit requirements are unresolved.

Quality Lens — Publishing and Scheduling Across Platforms

Failure signal

- A retry produces two uploads or exposes an unreviewed caption.

Repair move

- Gate the action with approval plus an idempotency key and reconcile the returned post ID.

Quality evidence

- The release log links one approved package to one intended platform action.

Analysing Performance

WHAT

- Performance analysis starts with a decision and a metric contract. Reach, starts, watch time, average view duration, completion, saves, shares, comments, and downstream actions describe different parts of audience response. The agent should calculate defined metrics at a consistent grain, compare appropriate windows, and separate observation from explanation.

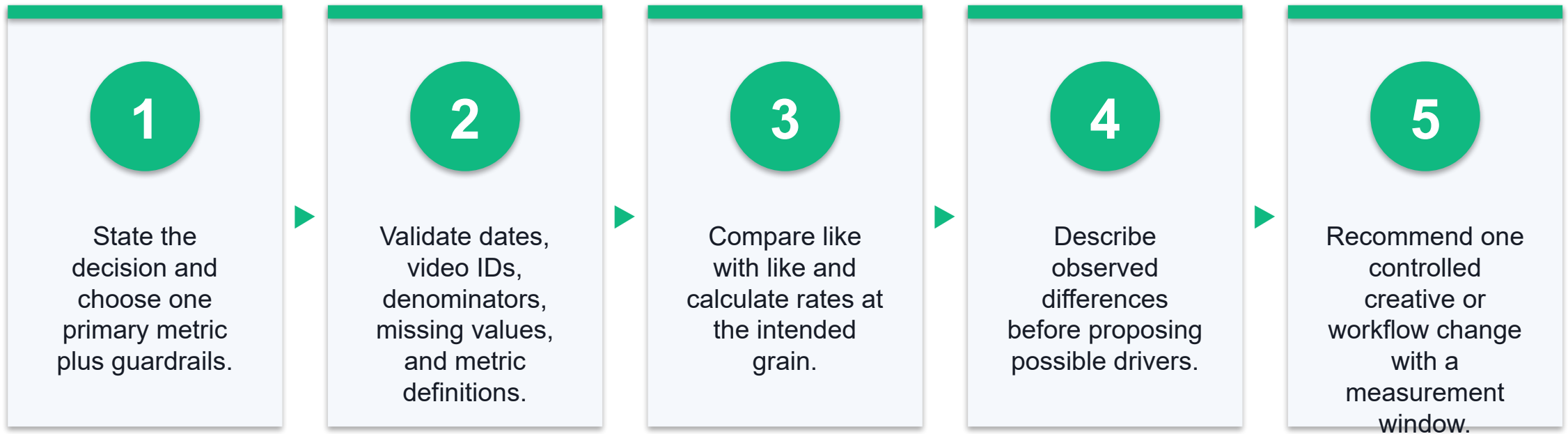
WHY IT MATTERS

- A large view count does not prove that a creative choice caused success. Platform metric definitions and counting rules can change, and small samples are unstable. A useful analysis agent preserves denominators, dates, segment, content version, and data source, then proposes a limited next test instead of declaring a universal rule.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Analysing Performance



In Practice — Analysing Performance

WORKED EXAMPLE

- The Harbour Bean synthetic dataset compares eight posts by hook family and duration.
- The agent calculates completion and save rates from the supplied counts, then flags low-impression rows.
- It recommends testing the strongest clear-promise hook while holding topic and duration band stable.

DECISION GUIDE

- **Use when**
 - The data source, metric definitions, window, and comparison grain are known.
 - A proposed action can be tested and reviewed against guardrails.
- **Avoid when**
 - Denominators are missing or metrics from different platforms are treated as identical.
 - A single high-performing post is being treated as causal proof.

Quality Lens — Analysing Performance

Failure signal

- The dashboard ranks videos by a rate calculated from incompatible denominators.

Repair move

- Write the metric contract and validate each row before analysis.

Quality evidence

- Every recommendation cites a defined metric, segment, window, caveat, and next test.

Scaling Your Video Content Pipeline

WHAT

- Scaling means increasing useful throughput without losing evidence, control, or quality. The team standardises production contracts, templates, reusable agent tools, asset naming, review rubrics, and telemetry. Work is queued and prioritised; capacity and cost budgets are visible; exceptions are handled explicitly.

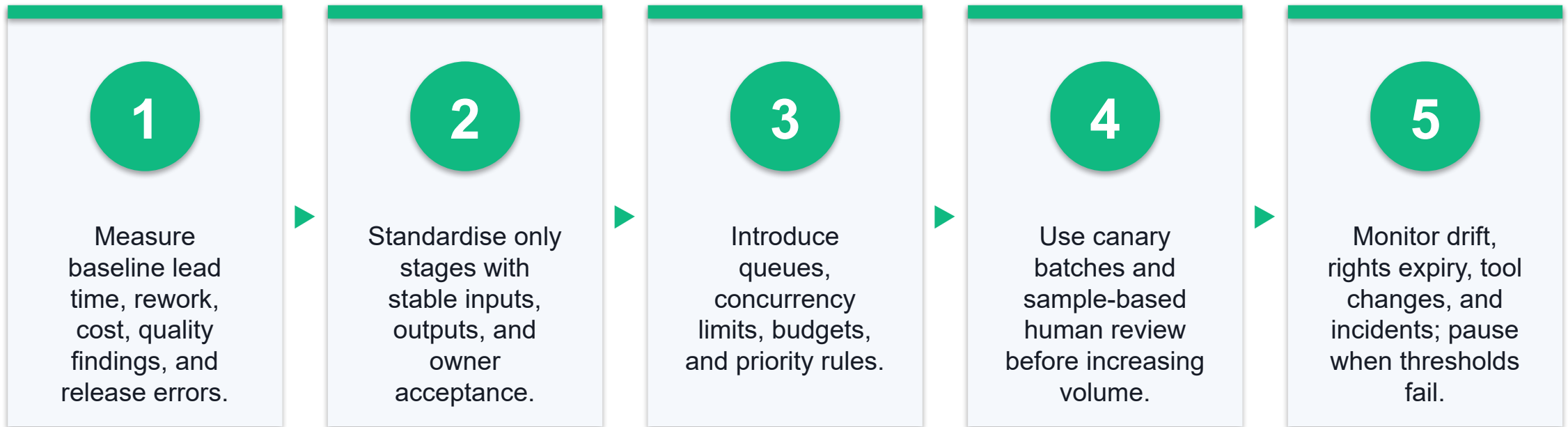
WHY IT MATTERS

- Volume magnifies small defects. A weak prompt creates many weak scripts, a permissive credential multiplies risk, and a missing rights record blocks an entire catalogue. Scale should follow demonstrated reliability at lower volume. Versioned templates, sampling, canary releases, and stop conditions keep growth reversible.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Scaling Your Video Content Pipeline



In Practice — Scaling Your Video Content Pipeline

WORKED EXAMPLE

- The team moves from one video to a three-video weekly batch using the same approved production contract.
- Generation is capped per run, every fifth draft receives an additional cold review, and public release stays human-approved.
- A scorecard tracks cycle time, cost per accepted video, rework rate, blocked rights items, and post-release learning.

DECISION GUIDE

- **Use when**
 - The pilot is reliable, measurable, and recoverable.
 - Owners, limits, incident response, and manual fallback are in place.
- **Avoid when**
 - Quality evidence is incomplete or rework already consumes more time than the workflow saves.
 - The plan increases permissions or public actions faster than monitoring and review capacity.

Quality Lens — Scaling Your Video Content Pipeline

Failure signal

- Output volume rises while accepted-video cost and rework also rise.

Repair move

- Throttle the queue and fix the earliest stage producing repeated defects.

Quality evidence

- Throughput improves while guardrail metrics remain within agreed limits.

Hands-On Labs — Automating and Scaling Video Production

Lab 7

- Orchestrate the Private Release Package with Human Approval

Lab 8

- Analyse Synthetic Performance and Build the Scaling Control Plan

Orchestrate the Private Release Package with Human Approval

LAB 7

You will import a release orchestrator, inspect its state transitions, validate video, caption, metadata, rights, and disclosure fields, then exercise denial and approval paths. The lab creates a dry-run package only; no public account or live posting credential is used.

You'll build

A release-package.json, denied-approval.json, private-release-approval.json, and release-orchestrator-reviewed.json in stage folder 04-release.

1 Import the orchestrator

2 Validate release inputs

3 Exercise denial

4 Issue scoped approval

Tools: n8n, supplied release workflow, reviewed video package

Orchestrate the Private Release Package with Human Approval

Test it

The validator must print LAB-07 PASS. It recomputes the canonical package and every video/caption/rights/final-review hash, requires an exact approval hash plus future expiry, and validates zero-action platform previews.

Analyse Synthetic Performance and Build the Scaling Control Plan

LAB 8

You will import a deterministic analytics workflow, validate metric definitions and denominators, calculate completion and save rates, inspect low-volume caveats, and produce one creative experiment. You will then complete a scaling scorecard covering throughput, cost, rework, quality, rights, incidents, and rollback.

You'll build

A metric-contract.json, analytics-result.json, next-test.md, scaling-scorecard.csv, and integrated-handover.md in stage folder 05-learn.

1 Define metric contracts

2 Validate synthetic data

3 Calculate and compare

4 Design one next test

Tools: n8n, supplied synthetic analytics CSV, spreadsheet or text editor

Analyse Synthetic Performance and Build the Scaling Control Plan

Test it

The validator must print LAB-08 PASS. It proves low-volume exclusion, baseline/pilot, duplicate-release, rollback, and capacity gates, exact scorecard controls, the complete next test/handover, and a structured HOLD/canary decision.

Recap — Automating and Scaling Video Production

1

You can now: LO4: orchestrate multi-step agents and prepare safe platform-specific publishing actions

2

You can now: LO4: analyse performance with metric contracts and scale the pipeline through measurable guardrails



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Design the Agent

Explain agentic AI, select suitable video-production tools, write bounded prompts, and design a controlled end-to-end workflow.

2

Generate the Assets

Build AI-assisted research, scripting, storyboard, visual, voiceover, and music hand-offs from an approved creative brief.

3

Edit and Review

Assemble and refine a short-form video through automated editing, captioning, branding, and evidence-based quality checks.

4

Publish and Scale

Orchestrate multi-step video agents with human approval, prepare safe publishing actions, analyse performance, and plan responsible scale.

Continue Building Safely

1

Replace one mock stage at a time and retain its contract and fallback.

2

Create test jobs for missing inputs, malformed output, tool failure, and denied approval.

3

Keep publishing private until platform permissions and an accountable release owner are confirmed.

4

Measure accepted output, rework, cost, latency, and incidents before increasing volume.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Keep your downloaded C436 course files and use the lab checkpoints to restart safely.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

Thank You!

You can now design and operate a bounded video-production agent from approved brief to evidence-led scaling plan.